PREMIUM STANDALONE TEXTBOOK

Machine Tools

Complete diploma-level handbook for metal cutting, lathes, drilling, shaping, planing, slotting, milling, grinding, superfinishing, NC/CNC machines, cutting fluids and lubricants.

Course Code: 3023 | Semester: 3 | Credits: 4 | 60 periods | Program Core

Start Learning

Four-module learning map

1. Metal Cutting & Lathe

Chip formation, tool nomenclature, Taylor equation, lathe construction and operations.

2. Drilling, Shaper, Planer, Slotter

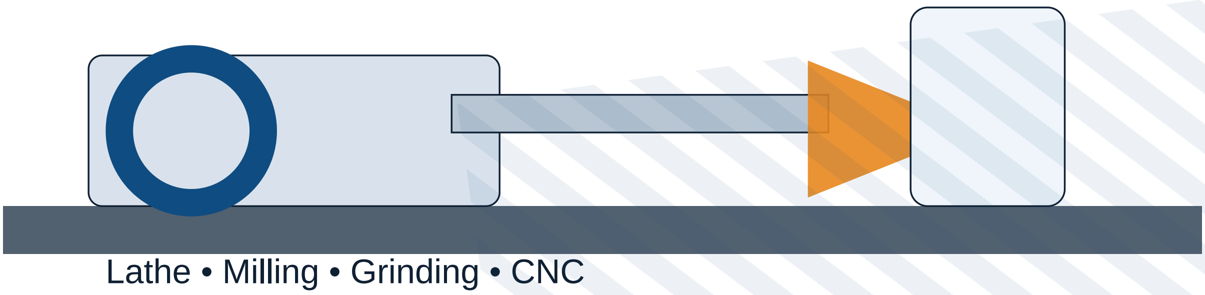
Drilling operations, quick return mechanisms and reciprocating machine tools.

3. Milling, Grinding & Finishing

Milling cutters, indexing, grinding wheels, centreless grinding, honing and lapping.

4. NC/CNC, Fluids & Lubricants

Machine control, axes, ATC, tool magazines, coolants and lubricant properties.



Course Overview

Program

Diploma in Mechanical Engineering / Manufacturing Technology.

Course category

Program Core; L:T:P = 3:1:0; periods per week = 4.

Purpose

Students learn how machine tools remove material safely and accurately to produce components.

Machine tools are power-driven production machines that hold a workpiece and a cutting tool in a controlled relationship. This handbook expands the official syllabus into a complete study guide with definitions, working principles, construction, operation sequences, formulas, solved examples, diagrams, practice questions and model answers.

Course Objectives and Prerequisites

Official objectives

1. To familiarize students with the concept and basic mechanics of metal cutting.
2. To familiarize students with the working of standard machine tools such as lathe, shaping, planing, milling, slotting, drilling and allied machine tools.
3. To introduce the basic concepts of NC and CNC machines.

Prerequisites

Topic	Course	Semester
Basic manufacturing operations performed in industries	Engineering Workshop Practice	1 and 2
Introduction to manufacturing processes	Manufacturing Technology	2

Workshop practice gives confidence with hand tools, work holding, marking, drilling and measurement. Manufacturing Technology introduces casting, forming and basic machining. These make it easier to understand why correct speed, feed, depth of cut, tool angles, rigidity, lubrication and safety are essential in machine tools.

Course Outcomes

CO	Outcome	Hours	Cognitive level
CO1	Describe mechanics of metal cutting and explain lathe machinery, parts and operations.	15	Understanding

CO2	Summarize drilling, shaper, planer and slotting machines including parts, operations and specifications.	14	Understanding
CO3	Describe milling and grinding machines, cutting tools, operations and superfinishing operations.	15	Understanding
CO4	Explain NC and CNC machines and the significance of lubricants and cutting fluids in machining.	14	Understanding
Series tests	Internal assessment tests	2	-

CO-PO Mapping Summary

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2		2				
CO2	2						
CO3	2						
CO4	2						2

Scale: 3 - strongly mapped, 2 - moderately mapped, 1 - weakly mapped.

Module 1: Metal Cutting and Lathe Machines

COURSE OUTCOME: CO1

Official module outcomes: M1.01 Compare orthogonal and oblique cutting; M1.02 identify single-point tool nomenclature and tool materials; M1.03 solve Taylor tool-life problems; M1.04 explain lathe classification, parts and specifications; M1.05 explain lathe operations.

Metal cutting fundamentals

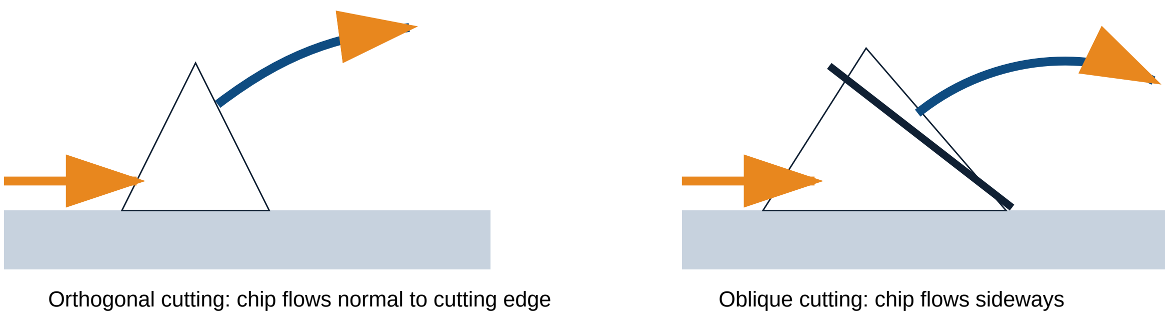
Machining removes unwanted material as chips using a wedge-shaped cutting tool. In the cutting zone the work material is compressed, sheared and separated. The primary deformation zone forms the chip, the secondary zone is at the tool-chip interface, and the tertiary zone is at the tool flank and machined surface.

Malayalam Support: □□□□□□□□□□□ □□□ □□□□□□□□□□□□ □□□□□□□ □□□□□□ □□□□□□□ □□□ □□□□□□□ □□□□□□□□□□□. □□□□□□□ □□□□□□□, □□□□, □□□ □□□□□□ □□□□□□□ □□□□□□□□□□□□ □□□□ □□□□□□□□ □□□□□□□ □□□ □□□□□□ □□□□□□□□□.

Orthogonal and oblique cutting

In orthogonal cutting the cutting edge is perpendicular to the direction of tool travel and chip flow is approximately normal to the edge. In oblique cutting the edge is inclined, so the chip flows sideways. Oblique cutting is common in actual lathe, milling and drilling operations because it gives smoother action and lower shock.

Malayalam Support: Orthogonal cutting-□ cutting edge □□□□ 90° □□□ □□□□□□□□□. Oblique cutting-□ edge □□□□□□□□□□□□□□□□□; □□□□□□□□□□ chip □□□□□□□□□□□ □□□□□□□.



Original inline SVG: cutting edge orientation and chip-flow direction.

Chip formation and chip types

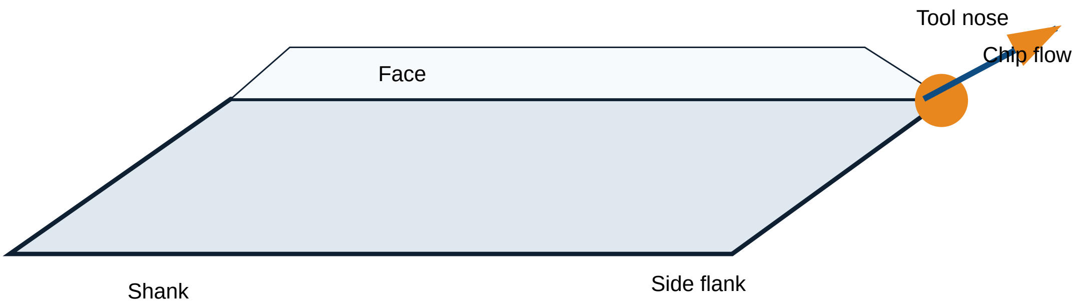
Continuous chips form in ductile materials at high speed and low feed; they give good finish but may entangle. Continuous chips with built-up edge occur when work material welds temporarily to the tool face. Discontinuous chips form in brittle materials, low speed or poor rigidity. Serrated chips occur in difficult-to-cut alloys because shear is cyclic.

Cutting parameters

Cutting speed is relative surface velocity between tool and work. **Feed** is tool movement per revolution or stroke. **Depth of cut** is the thickness of material removed normal to the machined surface. Material removal increases with speed, feed and depth of cut, but tool wear and heat also increase.

Single-point cutting tool nomenclature

A single-point tool has shank, face, flank, side flank, end flank, side cutting edge, end cutting edge and nose. Important angles are back rake, side rake, end relief, side relief, side cutting-edge angle, end cutting-edge angle and nose radius. The tool signature lists these angles in a standard order.



Cutting-tool materials

Carbon tool steel is inexpensive but loses hardness at high temperature. HSS has better hot hardness and toughness and is common for drills and taps. Cemented carbide permits high cutting speed and good wear resistance. Ceramics resist heat but are brittle. CBN and diamond are modern materials for hard and abrasive work materials.

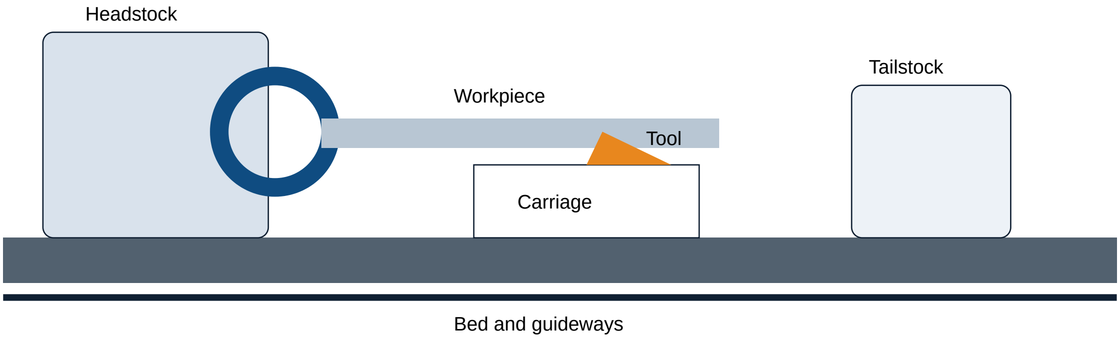
Tool wear, tool life and machinability

Flank wear occurs on the clearance face and affects dimensional accuracy. Crater wear occurs on the rake face due to hot chip flow. Tool life is the useful cutting time until a chosen wear limit. Machinability means how easily a material can be machined to acceptable finish, accuracy, tool life and power consumption.

Malayalam Support: Tool life □□□□□□□ □□□ □□□□□□□□□□□ □□□□ quality-□□□ machining □□□□□□□ □□□□□□□□□ □□□□. Cutting speed □□□□□□□□□□ tool life □□□□□□□ □□□□□□□.

Lathe machines

A lathe rotates the workpiece and feeds a single-point cutting tool to generate cylindrical, flat, tapered and threaded surfaces. Main parts are bed, headstock, spindle, speed gearbox, chuck, tailstock, carriage, saddle, cross slide, compound rest, tool post, apron, lead screw, feed rod and rack-and-pinion mechanism.



Lathe types and operations

Centre lathe is general purpose. Tool-room lathe is precise and versatile. Bench lathe is small and mounted on a bench. Speed lathe is simple and high speed for wood turning, polishing and spinning. Operations include facing, cylindrical turning, step turning, taper turning by compound rest, tailstock set-over, taper attachment and form tool, thread cutting using lead screw and change gears, drilling, boring, knurling and parting.

> **Malayalam Support:** Taper turning-□ diameter □□□□□□□□ □□□□; □□□ □□□□□□□□□ □□□□□□□□ □□□□□□ □□□□□□□□□ □□□□□□□□□□□□. Compound rest □□□□□□□□□ tailstock set-over □□□□□□□ taper □□□□□□□□□□□.

Important comparisons

Topic	Option A	Option B
Orthogonal vs oblique	Edge perpendicular, chip normal to edge, simpler analysis.	Edge inclined, chip flows sideways, common in practice.
HSS vs carbide	Tough, cheaper, lower speed.	High hot hardness, high speed, less tough.
Three-jaw vs four-jaw chuck	Self-centering, quick for round/hex work.	Independent jaws, accurate centering, irregular jobs.
Lead screw vs feed rod	Thread cutting with accurate pitch.	Power feed for general turning and facing.

High-priority practice

Definitions

Define the key process, list important parts and state applications.

Diagram

Draw and label the main machine or mechanism from this module.

Numerical / selection

Solve related formula problems or select a suitable machine, tool, fluid or process.

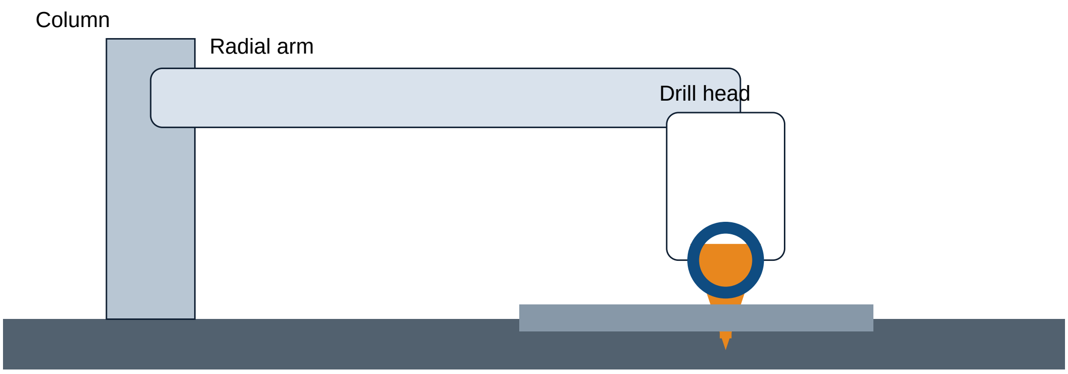
Module 2: Drilling, Shaping, Planning and Slotting Machines

COURSE OUTCOME: CO2

Official module outcomes: M2.01 shaper and planer classification, parts and mechanisms; M2.02 shaper versus planer; M2.03 drilling machines; M2.04 drills, reamers and operations; M2.05 slotting machine uses, parts and specifications.

Drilling machines

Drilling produces round holes using a rotating drill and axial feed. Portable, sensitive, upright, radial, gang and multi-spindle drilling machines are selected according to size, accuracy and production rate. A radial drilling machine has base, column, radial arm, drill head, spindle, table and feed mechanism, so heavy jobs can remain stationary while the drill head is positioned.



Drilling tools and operations

A twist drill has shank, body, flutes, land, margin, web, point, cutting lips, chisel edge, helix angle and point angle. Drilling starts a hole, reaming improves size and finish, boring enlarges an existing hole, counterboring forms a flat-bottom enlargement, countersinking forms a conical seat, spot facing creates a local flat surface, tapping cuts internal threads and centre drilling creates centre holes.

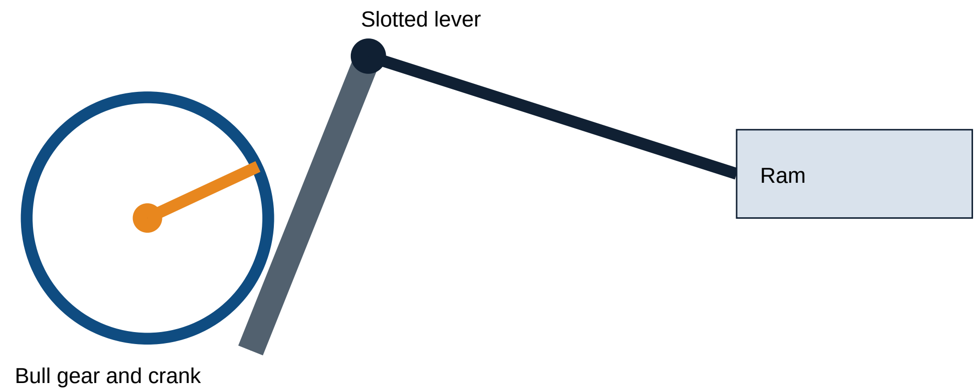
Shaping machines

A shaper uses a single-point tool mounted on a reciprocating ram. The tool cuts during forward stroke and returns idle during return stroke. Parts include base, column, ram, tool head, clapper box, table, cross rail, vice and tool post. Horizontal, vertical, standard and universal shapers are used for flat surfaces, grooves, slots and inclined surfaces.

Malayalam Support: Quick-return mechanism-□ cutting stroke □□□□□□□ slow □□□□ return stroke fast □□□□ □□□□□□□. □□□□□□□□□ idle time □□□□□□.

Crank and slotted lever quick-return mechanism

A rotating bull gear drives a crank pin that slides in a slotted lever. The lever oscillates and moves the ram through a link. Because the crank rotates through different angles during cutting and return strokes, the return stroke takes less time. Stroke length and stroke position are adjusted by changing crank radius and ram position.



Planing and slotting machines

A planer machines large workpieces. The work table reciprocates under one or more stationary tool heads. Main parts are bed, table, columns, cross rail, tool heads and drive mechanism. A slotter is a vertical shaper; its vertical ram cuts internal keyways, slots and internal profiles. Slotter parts are base, column, table, ram, tool head, rotary table, work-holding devices and tool-holding devices.

Important comparisons

Comparison	First	Second
Drilling vs boring vs reaming	Drilling creates a hole.	Boring enlarges; reaming finishes and sizes.
Shaper vs planer	Tool reciprocates; medium jobs.	Work table reciprocates; large jobs.
Shaper vs slotter	Horizontal ram for flat surfaces.	Vertical ram for internal slots/keyways.

High-priority practice

Definitions

Define the key process, list important parts and state applications.

Diagram

Draw and label the main machine or mechanism from this module.

Numerical / selection

Solve related formula problems or select a suitable machine, tool, fluid or process.

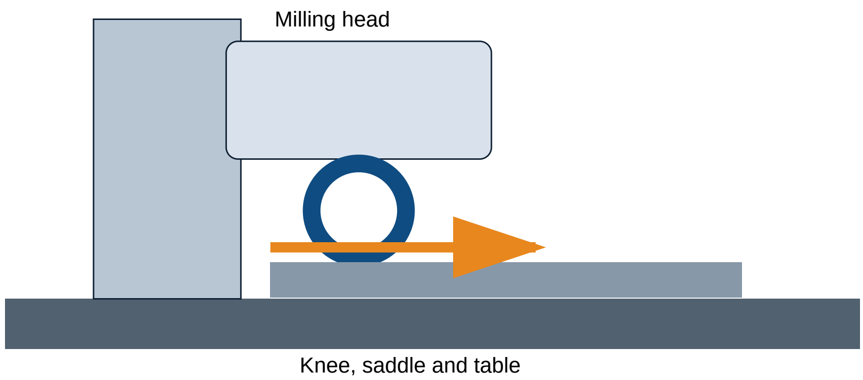
Module 3: Milling, Grinding and Superfinishing

COURSE OUTCOME: CO3

Official module outcomes: M3.01 milling process and specifications; M3.02 milling cutters; M3.03 milling tool and work holding; M3.04 grinding types, tools and material removal; M3.05 superfinishing processes.

Milling machines

Milling removes metal by a rotating multi-tooth cutter while the work is fed against it. Plain milling machines have horizontal spindle and are used for flat surfaces. Universal milling machines include a swivelling table and dividing head for helical and indexed work. Vertical milling machines have vertical spindle and are useful for end milling, face milling, slots and dies.



Up milling and down milling

In up milling the cutter rotation opposes feed at the point of contact; chip thickness starts at zero and increases. It is safer on machines with backlash but surface finish can be lower. In down milling cutter rotation is in the same direction as feed; chip thickness starts maximum and decreases, giving better finish but requiring backlash-free feed.

Malayalam Support: Up milling-□ cutter feed-□□ □□□□□□□□□□□□□. Down milling-□ cutter feed direction-□□□□□□ □□□□□□□□□□□□□□□□□□□□; backlash □□□□□□□□□□ down milling □□□□□□□□□□□□.

Indexing

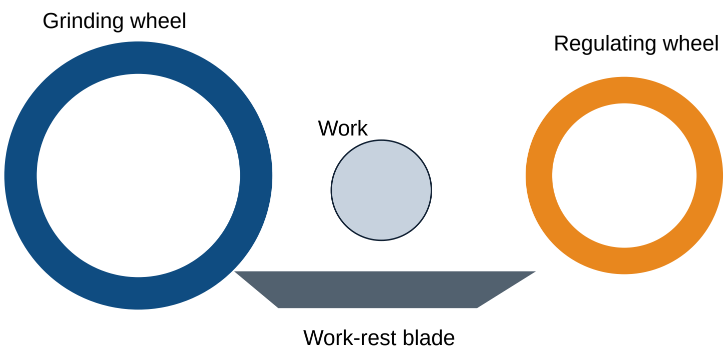
Indexing divides a circular workpiece into equal angular parts using a dividing head. For a standard 40:1 dividing head, crank turns for N divisions = 40/N. Simple indexing uses available index-plate holes. Compound indexing combines two crank movements. Differential indexing uses gear train motion when direct hole circles are not suitable.

Milling cutters and holding devices

Common cutters include slab mill, side-and-face cutter, end mill, face mill, angle cutter, form cutter, T-slot cutter, Woodruff keyseat cutter and gear cutter. Cutter tooth terms include rake angle, clearance angle, land, flute and cutting edge. Tool holders include arbor, collet and end-mill holder. Work holding uses machine vice, dividing head, rotary table, clamps and fixtures.

Grinding

Grinding uses bonded abrasive grains as many small cutting edges. A grinding wheel consists of abrasive grains, bond and pores. Natural abrasives include emery and corundum; artificial abrasives include aluminium oxide and silicon carbide. Bonds include vitrified, silicate, shellac, rubber and bakelite. Wheel selection depends on work material, stock removal, finish, area of contact and wheel speed.



Finishing processes

Centreless grinding supports work on a blade between grinding and regulating wheels; it has high production rate but needs careful setup. Honing uses bonded abrasive stones with rotation and reciprocation to produce cross-hatch patterns in cylinders. Lapping uses loose abrasive slurry between lap and work for high accuracy. Superfinishing uses low-pressure abrasive action to improve surface texture with minimal stock removal.

Important comparisons

Comparison	Key points
Plain vs universal vs vertical milling	Plain: horizontal spindle. Universal: swivelling table. Vertical: vertical spindle.
Up vs down milling	Up is safer with backlash; down gives better finish on rigid backlash-free machines.
Honing vs lapping vs superfinishing	Honing corrects bores; lapping gives high accuracy; superfinishing improves texture with low pressure.

High-priority practice

Definitions

Define the key process, list important parts and state applications.

Diagram

Draw and label the main machine or mechanism from this module.

Numerical / selection

Solve related formula problems or select a suitable machine, tool, fluid or process.

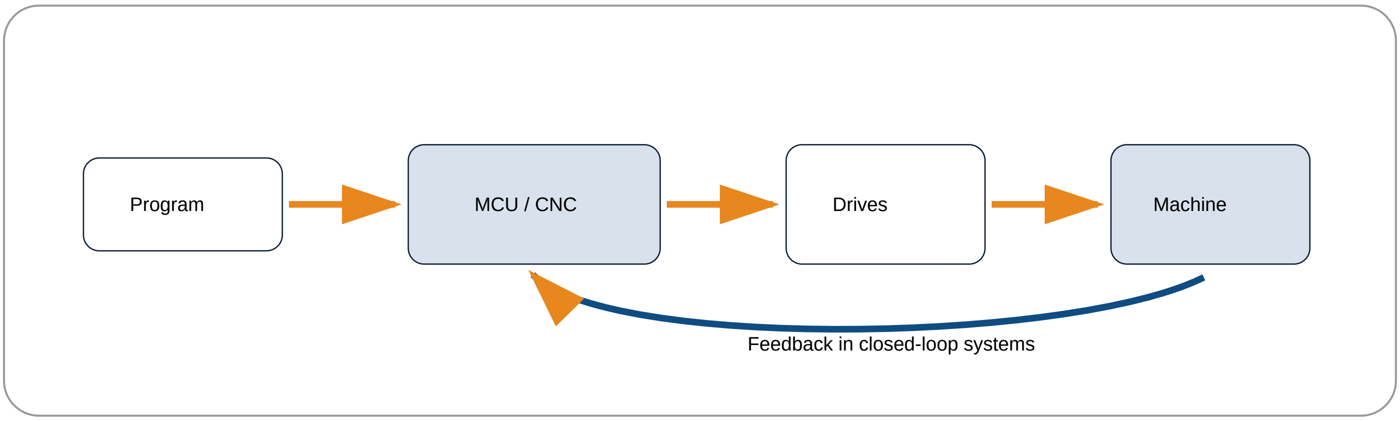
Module 4: NC/CNC Machines, Cutting Fluids and Lubricants

COURSE OUTCOME: CO4

Official module outcomes: M4.01 NC/CNC principles; M4.02 components, classification and tool systems; M4.03 compare NC and CNC; M4.04 cutting fluids and lubricants; M4.05 lubricant properties and applications.

NC and CNC fundamentals

Numerical control uses coded numerical instructions to control machine motions. Computer numerical control uses a computer-based controller with memory, editing, diagnostics and flexible control. Main blocks are input program, machine control unit, drive system, feedback system and machine tool.



Classification

By motion type: point-to-point, straight-cut and contouring control. By control loop: open-loop and closed-loop. By axes: two-axis, three-axis and multi-axis machines. Open-loop systems send commands without measuring actual output; closed-loop systems use encoders or feedback devices to correct position errors.

Malayalam Support: Closed-loop control-□ machine □□□□□□□□ □□□□□□□ feedback □□□ controller-□□□□□□□ □□□□□□□□□□ □□□□□□□□□□. □□□□□□□ □□□□□□□□□□□□ controller correction □□□□□□□□□□□.

Components and tool systems

CNC components include machine structure, control panel, display unit, spindle drive, servo motor, stepper motor, ball screw, linear guide, encoder, limit switches and lubrication/coolant systems. Automatic tool changer selects tools from a magazine and exchanges them with the spindle or turret. Tool magazines may be carousel or chain type.

Cutting fluids

Cutting fluids cool the cutting zone, lubricate tool-chip and tool-work interfaces, flush chips, improve tool life, improve surface finish, reduce cutting force and protect against corrosion. Types are straight cutting oils, soluble oils, synthetic fluids and semi-synthetic fluids. Application methods include flood, mist, jet, manual application and minimum-quantity lubrication.

Lubricants

Lubricants reduce friction and wear, remove heat, seal moving parts and prevent corrosion. They may be solid, liquid or gaseous. Important properties are viscosity, viscosity index, flash point, fire point, cloud point, pour point, oiliness, emulsification and oxidation stability. Machine tools use lubricants in bearings, gears, lead screws, slideways, ball screws, hydraulic systems and spindles.

Important comparisons

Comparison	Option A	Option B
NC vs CNC	Harder to edit, lower flexibility.	Computer controlled, editable, diagnostics and memory.
Open loop vs closed loop	No feedback, simple, cheaper.	Feedback correction, accurate, costlier.
Cooling vs lubrication	Removes heat from cutting zone.	Reduces friction at interfaces.

High-priority practice

Definitions

Define the key process, list important parts and state applications.

Diagram

Draw and label the main machine or mechanism from this module.

Numerical / selection

Solve related formula problems or select a suitable machine, tool, fluid or process.

Machine Tool Identification, Accessory and Diagram Bank

Centre lathe

Long bed, headstock, tailstock and carriage.

Radial drilling machine

Column, radial arm and movable drill head.

Shaper

Reciprocating ram and clapper box.

Planer

Large reciprocating table and bridge columns.

Slotter

Vertical ram for internal slots.

Vertical milling machine

Vertical spindle and table feeds.

Surface grinder

Grinding wheel, magnetic chuck and table.

CNC turning centre

Enclosed machine with control panel and turret.

Photo placeholders are intentionally replaced by original offline identification cards and SVG-style diagrams to avoid unverified copyrighted images. Use these as machine identification notes.

Formula Bank with Solved Examples

Cutting speed

$V = \pi DN / 1000$ for turning, drilling and milling when D is in mm and N is rpm. V is in m/min.

Example: D = 50 mm, N = 400 rpm. $V = \pi \times 50 \times 400/1000 = 62.83$ m/min. **Common mistake:** forgetting division by 1000.

Spindle speed

$N = 1000V / \pi D$. Use when recommended cutting speed and diameter are known.

For V = 30 m/min, D = 20 mm, $N = 1000 \times 30/(\pi \times 20) = 477$ rpm. Select nearest safe machine speed.

Depth of cut

$d = (D1 - D2)/2$. Turning removes material from diameter on both sides.

D1 = 60 mm, D2 = 54 mm. d = (60-54)/2 = 3 mm.

Feed rate and time

Feed rate = fN, **T = L/(fN)**. f is mm/rev and L is effective travel in mm.

L=120 mm, f=.25 mm/rev, N=300 rpm. T=120/(.25×300)=1.6 min.

Taylor tool-life equation

$VT^n = C$. Higher speed generally means lower tool life.

If n=.25 and C=300, at V=150 m/min: T=(300/150)^(1/.25)=2^4=16 min.

Taper

Taper = (D-d)/L; **tan α = (D-d)/(2L)**.

D=40, d=30, L=100: taper=.1; tan α=.05 so α≈2.86°.

Milling table feed

F = fz Z N. fz = feed per tooth, Z = teeth, N = rpm.

fz=.08 mm/tooth, Z=10, N=250: F=200 mm/min.

Simple indexing

Crank turns = 40/N for a 40:1 dividing head.

For 20 divisions: 40/20 = 2 turns per division.

Grinding wheel speed

V = πDN / 60000. D in mm, N in rpm, V in m/s.

D=300 mm, N=1800 rpm: V=π×300×1800/60000=28.27 m/s.

Quick-return ratio

QRR = time of cutting stroke / time of return stroke. A value greater than 1 indicates faster return.

Cutting time 4 s, return time 2.5 s: QRR = 1.6.

Practice Section

Fill in the blanks

1. The chip flows sideways in ______ cutting.
2. The lead screw is mainly used for ______ cutting.
3. A ______ machine has a radial arm.
4. A standard dividing head ratio is ______.
5. In closed-loop CNC, ______ is used for correction.

MCQs

1. Best process for internal keyway: (a) milling (b) slotting (c) lapping (d) sawing.
2. Bond common in grinding wheels: (a) vitrified (b) wood (c) leather (d) wax.
3. Better finish on backlash-free milling machine: (a) down milling (b) up milling.
4. Tool material for high speed machining: (a) carbide (b) soft iron.

Numericals

1. Find N for V=40 m/min and D=80 mm.
2. Find turning time for L=150 mm, f=.3 mm/rev, N=250 rpm.
3. Find tool life when V=120, n=.25, C=300.
4. Find indexing crank turns for 16 divisions.

Quick Revision and Final Examination Checklist

Module 1

Orthogonal/oblique cutting, chip types, speed-feed-depth, tool angles, tool materials, Taylor equation, machinability, lathe parts, accessories, taper turning and thread cutting.

Module 2

Drilling machine classification, drill/reamer nomenclature, drilling operations, shaper parts, quick-return mechanism, shaper vs planer, slotter uses.

Module 3

Milling types, up/down milling, indexing, cutters, grinding wheel code, abrasives, bonds, centreless grinding, honing, lapping, superfinishing.

Module 4

NC/CNC comparison, control loops, axes, drives, ATC, tool magazines, cutting fluids, application methods and lubricant properties.

Common mistakes

Wrong unit conversion in cutting speed, using diameter instead of radius incorrectly, confusing shaper and planer, confusing drilling/boring/reaming, ignoring backlash in down milling.

Must-draw diagrams

Lathe, single-point tool, radial drilling machine, shaper quick return, milling machine, centreless grinding, CNC block diagram and ATC.

Model Question Paper - Machine Tools (3023)

Time: 3 hours. **Maximum marks:** 100. Answer fresh syllabus-based questions; diagrams should be neat and labeled.

Part A - Very short answer

1. Define cutting speed and feed.
2. Name two types of chips.
3. What is tool signature?
4. State the use of a lead screw.
5. Name two operations performed on drilling machines.
6. What is quick return motion?
7. Define simple indexing.
8. Name two artificial abrasives.
9. What is closed-loop control?
10. State two functions of cutting fluid.

Part B - Short answer

1. Compare orthogonal and oblique cutting.
2. Explain four important parts of a centre lathe.
3. Differentiate drilling, boring and reaming.
4. Explain crank-and-slotted-lever quick return mechanism.
5. Compare up milling and down milling.
6. Explain grinding wheel selection factors.
7. Compare NC and CNC machines.
8. List lubricant properties and applications.

Part C - Long answer / problems

1. Draw a labeled centre lathe and explain construction and operations.
2. A 50 mm workpiece is turned at 400 rpm. Find cutting speed. If length is 120 mm and feed is 0.25 mm/rev, find machining time.
3. Draw and explain radial drilling machine and twist-drill nomenclature.
4. Explain shaper, planer and slotter with comparison and applications.
5. Explain milling cutters, tool holding, work holding and simple indexing. Find crank movement for 25 divisions.
6. Explain grinding, centreless grinding, honing, lapping and superfinishing.
7. Draw CNC block diagram and explain open-loop and closed-loop control, axes, ATC and tool magazine.
8. Explain types, selection and application methods of cutting fluids and classify lubricants.

Complete Model Answers and Answer Key

Part A answers

1. Cutting speed is relative surface speed between tool and work; feed is tool movement per revolution, stroke or tooth. 2. Continuous and discontinuous chips. 3. Tool signature is standardized listing of tool angles and nose radius. 4. Lead screw is used for thread cutting. 5. Drilling and reaming. 6. Quick return makes return stroke faster than cutting stroke. 7. Simple indexing divides a workpiece using a dividing head and index plate. 8. Aluminium oxide and silicon carbide. 9. Closed-loop control uses feedback to correct motion. 10. Cooling and lubrication.

Selected short answers

Orthogonal vs oblique: Orthogonal cutting has cutting edge perpendicular to tool travel and chip flow normal to edge. Oblique cutting has inclined edge and side chip flow; it is smoother and common in practice.

Drilling, boring and reaming: Drilling creates a hole with a drill. Boring enlarges and corrects an existing hole with a single-point tool. Reaming improves accuracy and surface finish using a multi-edge reamer.

NC vs CNC: NC uses numerical instructions with limited flexibility. CNC uses computer control, memory, program editing, diagnostics, offsets and easier changes. CNC improves productivity, repeatability and flexibility but needs higher investment and skilled maintenance.

Numerical answers

Model paper Q2: $V = \pi DN/1000 = \pi \times 50 \times 400/1000 = 62.83$ m/min. $T = L/(fN) = 120/(0.25 \times 400) = 1.2$ min. Interpretation: the cut takes about 72 seconds, excluding approach and setting time.

Practice: N for V=40 and D=80 is 159 rpm. Turning time for L=150, f=.3, N=250 is 2 min. Tool life for V=120, n=.25, C=300 is (300/120)^4 = 39.06 min. Indexing for 16 divisions is 40/16 = 2.5 turns.

Long-answer guidance with full points

Centre lathe: Define lathe, state work rotates and tool feeds. Draw bed, headstock, spindle, chuck, tailstock, carriage, cross slide, compound rest, tool post, apron, lead screw and feed rod. Explain each function. Describe facing, straight turning, step turning, taper turning, thread cutting, drilling and boring. Conclude with specifications: swing, distance between centres, spindle bore, speed range and bed length.

CNC answer: Define NC and CNC. Draw program -> MCU -> drive -> machine; add feedback path for closed loop. Explain axes X, Y, Z; drives using servo/stepper motors and ball screws; encoders for feedback; tool offsets; ATC sequence: tool selection, magazine indexing, spindle orientation, tool unclamp, exchange, clamp and resume. Compare carousel and chain magazines.

Practice answer key

Fill blanks: oblique; thread; radial drilling; 40:1; feedback. MCQs: 1-b, 2-a, 3-a, 4-a.

References and Online Resources

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